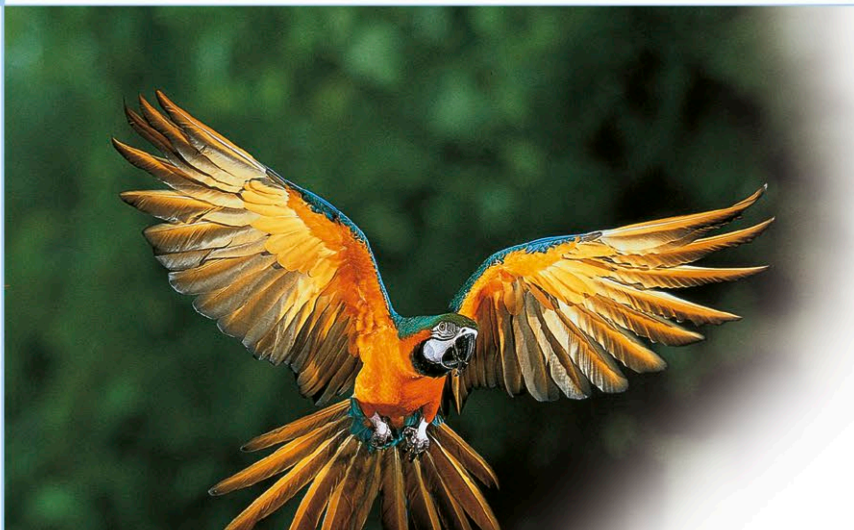


FAST FLYER
A blue and yellow macaw in flight displays its highly colorful plumage and long, elegant tail. The relatively narrow, tapering wings of macaws enable these birds to fly remarkably quickly, despite their large size.

Flight

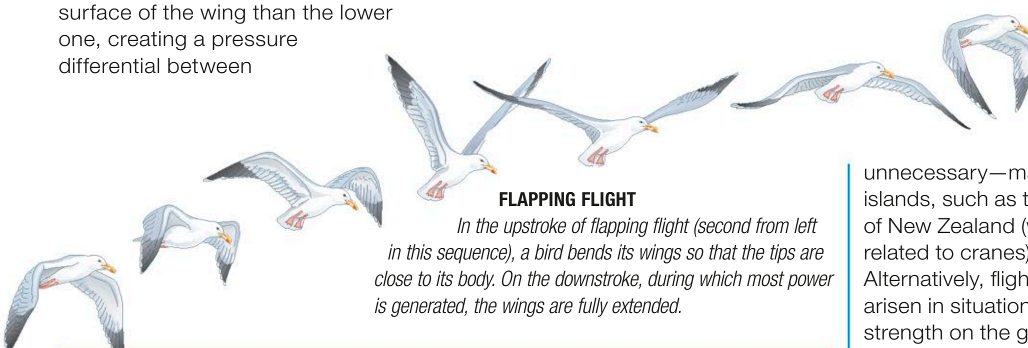
Birds are not the only animals capable of powered flight (bats and insects are similarly able), but they include among their number the largest and most powerful of all airborne animals.

A bird's wing achieves lift in the same way as does the wing of an aircraft. As the bird moves forward, air flows more quickly over the upper surface of the wing than the lower one, creating a pressure differential between the surfaces. Since it is the slower-moving air that exerts the greater pressure, the resulting force is upward. The strength of this upward force (called lift) increases with the size of the bird's wing and its forward speed (see panel, below).

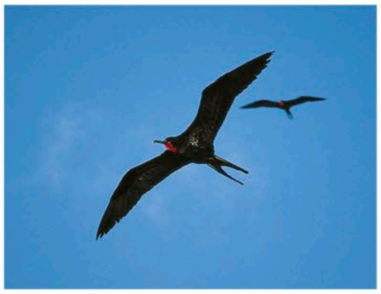
There are 2 main modes of flight: flapping and gliding (or soaring). Flapping does not create lift directly; instead, it generates the horizontal motion needed to increase airflow over the surface of the wing and produce lift. This is why most birds have to flap their wings to take off. When gliding, birds hold their wings outstretched and steady to the wind. A gliding bird consumes little energy, but it loses height as its speed, and hence the amount of lift, drop (unless it is riding a current of rising air).

Birds exhibit a great variety of wing shapes, reflecting the way they fly and their general lifestyle. Those that rely on a rapid takeoff and a short burst of speed to escape from predators tend to have broad, rounded wings, since these provide the required acceleration. Species that fly for long periods and need to save energy have long wings. Fast, powerful fliers, such as swifts and falcons, have long, curved wings with pointed tips to reduce drag. Tail shape is also important. Birds that make sudden changes of direction in midair, such as swallows and kites, often have forked tails.

Although most birds can fly, some are flightless—these include the ostrich, rheas, cassowaries, kiwis, the emu, and penguins. It is generally thought that these birds evolved from ancestors that once had the power of flight, since flightless birds have many adaptations for flight, such as hollow bones and wings (although the wings of many flightless birds have since become reduced in size). Flightlessness seems to have come about in one of several ways. For some birds in environments with relatively few predators, flight seems to have become unnecessary—many birds of isolated islands, such as the weka and takahes of New Zealand (which are closely related to cranes), are in this category. Alternatively, flightlessness may have arisen in situations where size and strength on the ground were more important for survival than flight. This is the case for the ostrich, which can run quickly, using its wings for balance; its long legs can also deliver a powerful kick to fend off predators. In penguins, the wings have become similar to flippers and are used by the bird to propel itself through water.



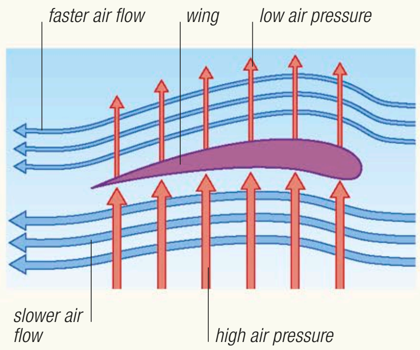
FLAPPING FLIGHT
In the upstroke of flapping flight (second from left in this sequence), a bird bends its wings so that the tips are close to its body. On the downstroke, during which most power is generated, the wings are fully extended.



AERIAL MANEUVERING
Frigatebirds are highly distinctive in outline, soaring and gliding above the open sea in search of food. Their narrow, angular wings enable them to fly with great speed and agility, and the characteristic long, forked tail helps them steer.

HOW BIRDS FLY

Birds have a large, keeled breastbone, to which massive flight muscles are attached. When the muscles contract, they bring about the powerful downstroke of the wing, which produces forward propulsion. When the muscles relax, the wing is pulled back up. Feathers (see p.264) also enable birds to fly efficiently. Those on the trailing edge of the wings and on the tail are designed to provide lift and aid maneuverability, while the visible body feathers (contour feathers) streamline the body in flight. In most birds, flying consists of wing-flapping or gliding. Several birds are capable of hovering in midair; hummingbirds can also fly backward.



AERODYNAMICS OF THE WING
A bird's wing is curved outward along the upper surface. As a result, air passing over it has to travel a slightly longer distance at a faster pace than the air traveling along the lower surface. The slow-moving air passing under the wing exerts greater pressure, producing an upward force known as lift.

Reproduction

Most birds are monogamous, breeding in pairs. In general, courtship consists of a male attracting a female with a visual display or by singing. Some birds perform spectacular displays: they may show off physical attributes, evident in the dazzling plumage displays of peacocks and birds of paradise; engage in dancing, as, for example, in cranes (see below); or make aerobatic flights, such as the dramatic swoops and dives of eagles. Many birds also draw attention to themselves using songs. The song, which is unique to the species, serves to repel male rivals and establish a breeding territory; it may also be used to attract a mate.

All birds reproduce by laying eggs, and they go to great lengths to ensure the survival of their clutch. Most do this by placing their egg (or eggs) in a specially built nest, which is either hidden or placed out of reach of predators. The eggs are incubated by one or more adults, almost always including the female. The number of eggs laid in a clutch varies greatly

COURTSHIP DANCE
A male and female red-crowned crane take part in their elaborate dance at the onset of the breeding season. One or both birds bow or bob their heads and leap into the air.

BIRDS